

The value of a primitive recursive term at the infinite point

a machine-checked computability theorem, from the trace alone

Abstract

Let f be a primitive recursive term, read as a monotone map on lazy naturals. We prove, constructively and by a machine-checked argument, that $f(S^\omega(\perp), \dots, S^\omega(\perp))$ is *computable*: a total function produces an explicit element of the domain, and that element is the least upper bound of the diagonal chain $f(S^m(\perp), \dots, S^m(\perp))$. The proof is carried out from a single invariant on the term's *trace* — the record of which variable the computation demands at each step and what it has produced — rather than from the ultimate-obstination property, which is not used anywhere and which we note does not follow from the trace invariant either. We then ask what survives under mutual recursion, where the ultimate-obstination property is outright false but the trace-level statement is not, and locate the one step that is still missing.

1 The domain, and primitive recursive terms

Let D be the domain of lazy naturals: its elements are $S^k(\perp)$ for $k \in \mathbb{N}$, $S^k(0)$ for $k \in \mathbb{N}$, and $S^\omega(\perp)$. The order is

$$S^j(\perp) \leq S^k(\perp) \iff j \leq k, \quad S^j(\perp) \leq S^k(0) \iff j \leq k, \quad S^j(\perp) \leq S^\omega(\perp),$$

and $S^k(0)$, $S^\omega(\perp)$ are maximal. In particular

$$S^k(0) \not\leq S^\omega(\perp), \tag{1}$$

the two maximal shapes being incomparable: a computation that has already produced a 0 is not an approximation of $S^\omega(\perp)$. Write $F \subset D$ for the *finite* elements, i.e. everything but $S^\omega(\perp)$. Say $x \in D$ is *incomplete* if it has the form $S^k(\perp)$ (so $S^\omega(\perp)$ is incomplete but not finite), *complete* if it is some $S^k(0)$, and write $\text{ht}(x) = k$ for the number of successors of a finite x . Note that a complete element is maximal among finite ones: if x is complete and $x \leq y$ with y finite, then $y = x$.

A primitive recursive term q of arity n is interpreted on finite tuples,

$$\text{ev}(q) : F^n \longrightarrow F,$$

by the evident clauses; $\text{ev}(q)$ is monotone. The extension to all of D^n is by continuity, and it is that extension we wish to *evaluate*.

2 The trace, and the invariant MP1

2.1 What a computation looks like from outside

Take a term $t(x_1, \dots, x_a)$, substitute $\perp$ for every variable, and ask for its normal form. Either it is already a numeral, or it is $S^{n_0}(u_0)$ where u_0 is stuck: it needs a variable, say x_{i_0} . Replace that

variable by $S(x_{i_0})$ — the least thing that could unstick it — and repeat. What comes out is a sequence

$$(n_0, i_0), (n_1, i_1), (n_2, i_2), \dots$$

where n_p is the number of successors produced after p substitutions and i_p is the variable needed next. Equivalently, recording how deep one has gone in each variable rather than which was taken last, it is a sequence

$$(n_p; a_{1p}, \dots, a_{ap}), \quad a_{1p} + \dots + a_{ap} = p.$$

This sequence has a *remarkable property*: it determines the value of t at every finite point at once. For

$$t(S^{l_1}(\perp), \dots, S^{l_a}(\perp)) \quad \text{has exactly } n_p \text{ successors, where } p \text{ is maximal with } a_{ip} \leq l_i \text{ for all } i.$$

One does not re-run the term at each point; one reads off the largest prefix of the sequence the point can pay for. Everything below is a consequence of that one observation, applied at increasingly complicated families of points.

2.2 The trace

We record the sequence, and a little more. A *trace* of arity a is either a leaf $\langle v \rangle$ carrying a value $v \in F$, or a node consisting of

- a *walk* $iv(0), iv(1), \dots$ with $iv(k) < a$ — the variable demanded at step k ;
- a *value sequence* $ov(0), ov(1), \dots$ in F — the value the term has when its replay sticks at step k ;
- for each coordinate $c < a$ and each $v \in \mathbb{N}$, a *continuation*, itself a trace of arity $a - 1$.

The walk generates its own levels: step k raises coordinate $iv(k)$ by one and leaves the others alone, so with

$$lv(c, 0) = 0, \quad lv(c, k+1) = \begin{cases} lv(c, k) + 1 & c = iv(k), \\ lv(c, k) & \text{otherwise,} \end{cases} \quad \sum_c lv(c, k) = k.$$

After k steps the walk stands at $(S^{lv(0,k)}(\perp), \dots)$, and $ov(k)$ is the value there: the pair (iv, ov) is exactly the sequence of §2.1, with $n_p = \text{ht}(ov(p))$ and $a_{ip} = lv(i, p)$. Recording the whole $ov(k)$ and not only its height is essential: a height cannot tell $S^k(0)$ from $S^k(\perp)$, and $\text{prec}(\text{zerf}, \text{zerf})$ — which is $\perp$ at $\perp$ and the numeral 0 above it — refutes any height-only version.

Replay. A trace is evaluated at an actual tuple X of finite elements by running the walk against the heights X has. Let

$$\text{stick}(X) = \text{the first step } k \text{ at which the walk asks for a level that } X \text{ does not have.}$$

This is a bounded search: the walk spends one level per step, so $1 + \sum_c \text{ht}(X(c))$ steps of fuel suffice. Writing $k = \text{stick}(X)$, $i = iv(k)$ and $v = \text{ht}(X(i))$, the value is given by a three-way case:

$$\text{sem}\langle v \rangle(X) = v, \quad \text{sem}(\text{node})(X) = \begin{cases} ov(k) & ov(k) \text{ complete,} \\ ov(k) & X(i) \text{ incomplete,} \\ \text{sem}(\text{continuation at } i, v) (X \text{ with } i \text{ deleted}) & X(i) \text{ complete.} \end{cases}$$

The three clauses say: the computation has already answered; or it is blocked on a coordinate that is still $S^j(\perp)$, and $ov(k)$ is the answer; or that coordinate has turned out to be a numeral, in which case it is frozen into the term and the computation continues at arity $a - 1$. The last clause is why a trace is a tree, and why every argument below which involves completeness is a recursion on the *arity*.

Two consequences are used constantly. First, *if no coordinate of X is complete the third clause never fires*, so

$$\text{sem}(T)(X) = ov(\text{stick}(X)) \quad (2)$$

— the whole semantics collapses to a single lookup in the value sequence. Second, since a walk spends at most one level per step, $lv(c, k) \leq k$; combined with the fact that it sticks exactly when the level it wants is the one the coordinate does not have, this gives the dichotomy of §2.4.

The trace of a term. The trace is built by induction on the term, and — this is the point of building it rather than postulating its shape — it is proved to *denote* that term: $\text{sem}(T_q)(X) = \text{ev}(q)(X)$ for every X of the right length. So the trace is not a separate model to be related to $\text{ev}(q)$ afterwards; it computes $\text{ev}(q)$, and iv, ov are a decomposition of that computation.

Remark 1 (provenance, and what this trace is *not*). The sequence (n_p, i_p) of §2.1, together with the observation that it determines the value of the term at every finite point at once, is the notion of trace of an email from Coquand to Colson of 1991; the direct constructive proof of Colson’s ultimate obstination theorem that came out of it is *Une preuve directe du théorème d’ultime obstination*, C. R. Acad. Sci. Sér. I **314** (1992). The trace of §2.2 is that notion, with the whole value $ov(k)$ recorded in place of its height alone, and with continuations added for the coordinates that go complete.

It is *not* the notion of trace of R. David (*On the asymptotic behaviour of primitive recursive algorithms*, Theoret. Comput. Sci. **266** (2001) 159–193, and *Decidability results for primitive recursive algorithms*, Theoret. Comput. Sci. **300** (2003) 477–504, the paper that extends the constructive result to mutual recursion), although the two are aimed at the same theorem. There a trace is a pair (e, λ) : an element e of the domain *together with a labelling* assigning to each accessible cell of e a word of tokens x_n , recording which cell of which argument was used to produce that cell. A term denotes a continuous map from traces to traces, the arguments being themselves traces. So David’s trace is a computation *at a given tuple of arguments*, decorated with provenance — in his own reading, a branch of a Berry–Curien sequential algorithm.

The trace used here carries no provenance and is attached to the term alone. It is the record of the canonical increasing walk that starts at $(\perp, \dots, \perp)$ and raises, at each step, exactly the coordinate the computation is stuck on, together with the value produced at each step; the value at an arbitrary tuple is recovered by *replay* (§2.2), not by reading a label. The two therefore record different things, and neither is here defined from the other: David’s is the finer datum at a fixed point, since it says which cells were used, while the present one is the finer datum along the increasing family, since it keeps the whole sequence of values — distinguishing $S^k(0)$ from $S^k(\perp)$ — and descends into a continuation when a coordinate becomes a numeral. The distinction matters for §5: mutual recursion is exactly what David’s traces were built to handle, and the question asked there is whether *this* trace, with its value sequences and descents, ports to a block — not whether his does.

2.3 The invariant

Definition 1 (MP1). A value sequence ov has a *verdict* when

- either $ov(k)$ is complete for some k — and then, completeness being maximal and ov monotone, ov is constant from k on;
- or no $ov(k)$ is complete, and there is a threshold past which $ht \circ ov$ is either constant or strictly increasing.

A trace satisfies **MP1** when at every node of it

- (i) the walk iv is eventually constant, and
- (ii) the value sequence ov has a verdict,

and every continuation satisfies **MP1** as well.

Clause (i) is ultimate obstination read along the walk: from some step on the computation demands one variable and never changes its mind. Clause (ii) is the three-way shape of the value along that variable.

Two remarks on the exact form. First, the completeness disjunct has to be kept separate: it is the case where the computation answers, and no statement about heights can express it. Second, in the incomplete case one asks for *constant or strictly increasing*, not merely for the weaker dichotomy “bounded or unbounded”. The weaker one is what the theorem of §3 consumes, but it is not closed under recursion: for a step whose height may drop back, the orbit’s verdict is equivalent to the limited principle of omniscience. Constant-or-strictly-increasing repairs this, and one sees why in the recursion clause — for a strictly increasing step a single comparison decides the orbit, since $x \mapsto \varphi(x)$ either fixes a point or moves it upward for ever.

Theorem 1. *The trace of every primitive recursive term satisfies **MP1**.*

2.4 The engine: the stuck coordinate

Everything in the proof of Theorem 1 runs on one observation. Suppose a trace is fed a family $X_0 \leq X_1 \leq \dots$ of argument tuples and let $N_t = \text{stick}(X_t)$, which is monotone. At time t the replay is stuck at step N_t on the coordinate $d = iv(N_t)$, and *the level it needs there is exactly the height d offers*: the levels a walk has consumed on the way stay below what was available, and sticking says the one it wants is not. Hence:

if d never grows again, the replay is stuck at N_t for ever; if d grows, that is precisely the level it was waiting for, and the replay advances.

There is no third possibility. So N strictly increases until it freezes, and it freezes for good — and *which* of the two happens is decided by the regime of the stuck coordinate, which the induction hypothesis supplies. Deciding it is moreover a *bounded* search: if N has not frozen by the threshold of clause (i), then N has passed that threshold, so from there the stuck coordinate is the single eventual demand, whose regime is already known.

Given (2), this settles the value: if the replay freezes, the value is constant; if it strictly increases, the fed trace’s own verdict transfers along it verbatim. The remaining case is a coordinate going *complete*, and there the replay freezes too — a complete value is maximal, so the coordinate never moves again — which makes the continuation descended into a *fixed* smaller trace, and the whole argument recurses at arity $a - 1$. At most a descents happen.

2.5 Composition

Let $f(\vec{x}) = g(h_1(\vec{x}), \dots, h_p(\vec{x}))$. The arguments read the *same* variables, so they cannot be stepped independently: driving them one at a time charges a shared level twice and makes the composite stick too early. (Concretely, $f(x) = g(x, x)$ with g the identity in its first argument then answers $S(\perp)$ at $S^2(\perp)$.) The state must therefore be the composite's own level vector, with each argument *replayed against it* — which is exactly the remarkable property of §2.1 applied to each h_i at the current $(l_1, \dots, l_a)$. Sharing is then automatic: raising one level advances every argument that was waiting for it, at no extra cost.

The composite demands the level that the argument g is currently blocked on needs; so the selected argument advances by at least one replay step per stage. That *drive* is what carries an argument's own verdict, which is stated in its private replay depth, over to the composite's clock. Together with the fact that the composite's demand is eventually constant — say it waits on argument J for ever — the three cases of §2.4 apply with g as the fed trace and the argument values as the family:

- if J 's value settles, the composite's value is constant, and this needs nothing but agreement at the coordinate g is waiting on;
- if J 's value keeps growing, g 's replay strictly increases and g 's own verdict transfers;
- if J 's value becomes a numeral, g 's replay freezes, the continuation is fixed, and the argument recurses at smaller arity.

2.6 Primitive recursion

Let $f(\perp, \vec{y}) = \perp$ and $f(S^{j+1}(\perp), \vec{y}) = g(S^j(\perp), f(S^j(\perp), \vec{y}), \vec{y})$. Along f 's own walk, past the threshold of clause (i) exactly one coordinate grows, and there are two shapes.

The recursion argument grows. Then the parameters are frozen and f 's value sequence is the *chain*

$$V_0 = \perp, \quad V_{j+1} = g(S^j(\perp), V_j, \vec{y}).$$

While V_j is incomplete every coordinate of g 's argument tuple is still of the form $S^i(\perp)$, so (2) applies and

$$V_{j+1} = \text{ov}_g(N_j), \quad N_j = g\text{'s replay depth at depth } j.$$

This is the situation of §2.4, and here it has a sharp reading. Writing g 's sequence as $(n_p; x_p, y_p, z_p)$ — n_p successors after p steps, having gone x_p deep in the recursion argument, y_p in the recursive value, z_p in the parameters — the criterion is

$$f \text{ loops in its first argument if and only if there is } p \text{ with } y(p+1) = y(p) + 1 \text{ and } n_p \leq y_p,$$

and then f 's own sequence is ultimately $(n_p; x, z), (n_p; x+1, z), \dots$ — constant number of successors — while otherwise it is essentially g 's. The two halves of the criterion say exactly that step p demands the recursive value and that the successors produced do not exceed the depth already consumed of it: that is, that g 's replay *stalls* on the recursive value.

Two facts make this decidable. First, *a stall is never on the recursion argument*: that coordinate grows by one per depth all by itself, whereas the levels consumed on the way to the sticking step are below what was available one depth earlier. Second, a stall on anything else is *permanent*: a parameter never moves again, and a stall pins the recursive value, so the coordinate it waits on does

not move either. Hence either the replay strictly increases at every depth — and g 's verdict transfers — or it freezes and the height is constant. Which one is settled by a bounded search, exactly as in §2.4, and here clause (i) for g is what bounds it: past g 's own threshold the stalled coordinate must be g 's eventual demand, and each of the three possibilities for that demand is decided.

A parameter grows. Then the recursion depth is frozen at some c , and f along the growing parameter is g applied c times over. An induction on c does it: layer $c + 1$ is g fed the family

$$(S^c(\perp), \text{layer } c, S^{l+t}(\perp), \dots),$$

in which the recursion argument and the frozen parameters never move, the growing parameter grows at every step, and the middle coordinate is whatever the induction hypothesis says. That is precisely the hypothesis of the general lemma of §2.4, so each layer is one application of it.

Remark 2. The direction is worth stating explicitly, because an earlier version of this development went the other way. The trace argument above does *not* consume the ultimate-obstination property; it is proved from the trace by induction on the term, and the ultimate-obstination property is not used anywhere in it.

3 The theorem

Fix an arity n and write

$$\Delta_m = (S^m(\perp), \dots, S^m(\perp)) \in F^n, \quad \Delta_\infty = (S^\omega(\perp), \dots, S^\omega(\perp)) \in D^n.$$

The chain $c_m := \text{ev}(q)(\Delta_m)$ is monotone in m by monotonicity of $\text{ev}(q)$.

Theorem 2. *For every primitive recursive q of arity n one can compute an element $\text{val}(q) \in D$ which is the least upper bound of $(c_m)_{m \in \mathbb{N}}$. In particular $q \mapsto \text{val}(q)$ is total, so $f(S^\omega(\perp), \dots, S^\omega(\perp))$ is computable, uniformly in q .*

The all-infinite point is the *easy* point, and the proof is three lines of the machinery of §2.

Lemma 1 (the chain is a lookup). $c_m = \text{ov}(N_m)$, where N_m is the replay depth of the trace against the level vector $(m, \dots, m)$.

Proof. No coordinate of Δ_m is complete, so (2) applies; and the trace denotes $\text{ev}(q)$. □

Lemma 2 (the lookup is cofinal). $N_m \geq m$.

Proof. The walk spends at most one level per step, so at step $k < m$ it has consumed at most $k < m$ levels of the coordinate it is about to demand, and m are available. Hence it does not stick before step m . □

So the chain, reindexed, runs through the value sequence cofinally, and its least upper bound is the least upper bound of ov . The verdict of Definition 1 names that:

$$\text{val}(q) = \begin{cases} \text{ov}(k) & \text{ov complete at } k \text{ (a numeral),} \\ S^{\text{ht}(\text{ov}(k))}(\perp) & \text{ov never complete, height constant from } k, \\ S^\omega(\perp) & \text{ov never complete, height strictly increasing from } k. \end{cases}$$

Proof of Theorem 2. Complete at k . Then ov is constant from k on. For any m , $ov(N_m) \leq ov(k)$: either $N_m \leq k$ and this is monotonicity, or $N_m \geq k$ and the two are equal. So $ov(k)$ is an upper bound, and by Lemma 2 it is attained at $m = k$, hence least.

Never complete, height constant from k . Every c_m is $S^{\text{ht}(ov(N_m))}(\perp)$, and $\text{ht}(ov(N_m)) \leq \text{ht}(ov(k))$ by the same split. Attained at $m = k$, again by Lemma 2.

Never complete, height strictly increasing from k . Each c_m is incomplete, hence below $S^\omega(\perp)$, so $S^\omega(\perp)$ is an upper bound. It is the least one: a strictly increasing height is unbounded, so for any finite d — of either shape — Lemma 2 produces an m with $\text{ht}(c_m) > \text{ht}(d)$, and d is not an upper bound. $\square$

Remark 3. Only Lemma 1 uses that *every* coordinate of the point is infinite, and it uses only that none is complete and none is finite — so nothing descends and nothing blocks. That is the whole reason the argument at Δ_∞ is so much shorter than at a general point.

Remark 4 (what MP1 does not give). It does *not* give the ultimate-obstination property. The leaf trace $\langle S^0(\perp) \rangle$ of arity one satisfies MP1 vacuously and denotes the constant $\perp$, whose ultimate-obstination property fails at the point (0): at an all-complete point the property leaves no room — its second case wants a coordinate that is incomplete and finite, its third wants one that is $S^\omega(\perp)$ — so its first case must hold and the value must be a numeral, which a constant $S^m(\perp)$ is not. The missing ingredient is *totality*, that a term applied to numerals returns a numeral; that is a separate (easy) induction on the term, and it is not needed for Theorem 2.

4 The formalisation

All of the above is formalised in Agda and checked with `--safe --without-K --exact-split`, with no postulate, no hole and no termination pragma.

- The trace, its semantics and its levels, and the proof that the trace of a term denotes the term: `TraceDef`, `ReplayLv`, `TrSat`, `TrDen`, `TrTerm`. The collapse (2) is `TrSat.sem-bot`.
- The invariant and its two clauses: `TrMP1`. Composition: `TrComp`, `TrCompNG`, `TrCompSel`, `TrSelStab` for the index clause, and `TrCompVal`, `TrCompVerdict`, `TrCompMP1` for the value clause. Recursion: `TrPrec`, `TrPrecIvPMP` for the index clause, and `TrPrecChain`, `TrPrecStall`, `TrPrecPhi`, `TrPrecParPhi`, `TrPrecOvP` for the value clause. The general lemma of §2.4 is `TrFeedR` (with `TrCompVerdict` as the special case in which the demand is known in advance to settle).
- Theorem 1 is `TrTermMP1`, and Theorem 2 is `PRInfMP1`. Neither uses the ultimate-obstination property; the only thing either takes from the development of that property is its well-formedness *predicate*.
- Remark 4 is `TrUOFail`, and the totality it refers to is `PRTot`.

The proof *runs*: a companion file checks by evaluation that

$$\begin{aligned} \text{zerf}(S^\omega(\perp)) &= 0, \\ \text{proj}_0(S^\omega(\perp)) = \text{succ}(S^\omega(\perp)) &= S^\omega(\perp), \\ E(S^\omega(\perp)) &= 0 \quad \text{where } E = \text{prec}(\text{zerf}, \text{zerf}). \end{aligned}$$

E is the informative one: $E(\perp) = \perp$ but $E(S^{m+1}(\perp)) = 0$, so the chain is $\perp, 0, 0, \dots$ and the least upper bound is a numeral, at a point that is infinite.

5 Does this generalise to mutual recursion?

Consider now a block of r functions defined by simultaneous primitive recursion,

$$f_j(S(x), \vec{y}) = h_j(x, f_1(x, \vec{y}), \dots, f_r(x, \vec{y}), \vec{y}), \quad 1 \leq j \leq r, \quad (3)$$

with the h_j already given. Which parts of the argument survive?

Why the question is now better posed. The ultimate-obstination property is *false* for blocks: the formalisation contains a two-function block whose value along the diagonal has height $\lfloor m/2 \rfloor$, which is neither constant nor strictly increasing, so the shape its third case demands cannot be met. That used to be an obstacle to porting Theorem 2, because the theorem was proved through that property. It no longer is: §3 uses only the trace and MP1, and $\lfloor m/2 \rfloor$ is unbounded, so the value at Δ_∞ is $S^\omega(\perp)$ and is computed correctly. What has to be ported is MP1, not the ultimate-obstination property.

What survives verbatim. Sections 2.4–2.6 are about a trace fed a monotone family; nothing in them mentions how the function was defined. In particular the stuck-coordinate dichotomy, the bounded search that decides it, and the descent at a coordinate that goes complete all apply to a block’s step terms as they stand. Lemmas 1 and 2 are facts about the point Δ_∞ and the order on D , and are untouched.

What is missing, and where. For a block one wants the analogue of MP1 for the tuple $(f_1, \dots, f_r)$, from MP1 for the h_j . Two things are known.

First, the *invariant* does not transfer for arbitrary traces: there is a block whose two step traces satisfy the corresponding hypotheses and for which the bare bounded-or-unbounded dichotomy implies the limited principle of omniscience. Classically the same point reads: “is the height of $f(S^m(\perp))$ bounded in m ?” is Π_1^0 -hard. But that refutation does *not* transfer to real step terms. Its step height is $b(n) + n$ for an arbitrary binary b , and what makes the orbit hard is exactly that b may drop from 1 to 0; a step term’s height is constant or strictly increasing past a threshold the proof produces, and $b(n) + n$ admits such a threshold only when b never drops after it — which already decides the principle. So the refutation kills the closure of the invariant over *arbitrary* traces; it exhibits no primitive recursive block whose value at Δ_∞ is uncomputable.

Second, and this is the actual obstacle, the block’s own trace in the present development is *height-only*: it records how tall each component is at each unfolding, not what it is. That is enough while no component answers, and on that cone the block’s height dynamics is proved to be the real block’s. It is wrong as soon as one component becomes a numeral, and the formalisation contains a two-line instance: with $f_1 = 0$ and $f_2 = f_1 + S(f_2)$, the real f_2 climbs by one per unfolding — so its value is $S^\omega(\perp)$ — while the height-only dynamics freezes it at $\perp$. The reason is exactly the third clause of the semantics of §2.2: a coordinate that has gone complete makes the computation *descend into a continuation*, and a height cannot see that.

What to do. Rebuild the block trace on value sequences and continuations, as the traces of §2.2 already are, rather than on heights. This is not a new idea but a transcription: the composite and the recursion are exactly “a trace fed a monotone family, with descents”, and §2.4–2.6 — in the formalisation, **TrCompVal**, **TrCompVerdict** and **TrFeedR** — carry that argument for single terms in the form a block needs. The walk itself does not have to change: the replay depth is computed from heights alone in the single-term case too. Only the value law does.

What is certain either way. The level-by-level verdict holds for blocks unconditionally: for every K one can decide whether the value reaches S^K , because a block component's height past its threshold is a deterministic monotone iteration, decided by $K + 1$ steps. So for blocks the value at the infinite point is in any case computable *as a lazy natural*, which is the constructively correct reading of such an object. Whether one can moreover produce it as an element of the three-constructor datatype D — which builds the bounded-or-unbounded dichotomy in, $S^k(\perp)$ and $S^\omega(\perp)$ being different constructors — is what the rebuilt trace would settle.